

Poročilo: FLoCIC kompresija v projektu Pametni Paketnik

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Opis integracije

Algoritem **FLoCIC** je bil integriran v sistem za prepoznavo obrazov (**ORV**) projekta Pametni Paketnik. Kompresija se uporablja za **arhiviranje sivinskih slik obrazov** uporabnikov ob registraciji.

Lokacija implementacije

Datoteka

`compression/flocic.py`

`compression/image_compressor.py`

`compression/test.py`

`ORV/face_recognition_api.py`

Opis

FLoCIC algoritem (JPEG-LS + interpolativno kodiranje)

Wrapper za integracijo z face recognition

Testiranje kompresije na vzorcnih slikah

Integracija v REST API

Zaslonske slike

1. FLoCIC algoritem - interpolativno kodiranje

```

flocic.py x
... compression > flocic.py
103
104 def predict(pixels, x, y, width, height):
105     if y == 0 and x == 0:
106         return pixels[0][0]
107     elif y == 0:
108         return pixels[0][x - 1]
109     elif x == 0:
110         return pixels[y - 1][0]
111     else:
112         left = pixels[y][x - 1]
113         top = pixels[y - 1][x]
114         diag = pixels[y - 1][x - 1]
115
116         if diag >= max(left, top):
117             return min(left, top)
118         elif diag <= min(left, top):
119             return max(left, top)
120         else:
121             return left + top - diag
122
123 def jpeg_ls_predict(pixels, width, height):
124     E = []
125
126     for y in range(height):
127         for x in range(width):
128             if y == 0 and x == 0:
129                 E.append(pixels[0][0])
130             elif y == 0:
131                 E.append(pixels[0][x - 1] - pixels[0][x])
132             elif x == 0:
133                 E.append(pixels[y - 1][0] - pixels[y][0])
134             else:
135                 pred = predict(pixels, x, y, width, height)
136                 E.append(pred - pixels[y][x])
137
138     return E
139
140 def interleave(E):
141     N = [E[0]]
142     for i in range(1, len(E)):
143         if E[i] >= 0:
144             N.append(2 * E[i])
145         else:
146             N.append(2 * abs(E[i]) - 1)
147     return N
148
149 def create_cumulative(N):
150     C = [N[0]]
151     for i in range(1, len(N)):

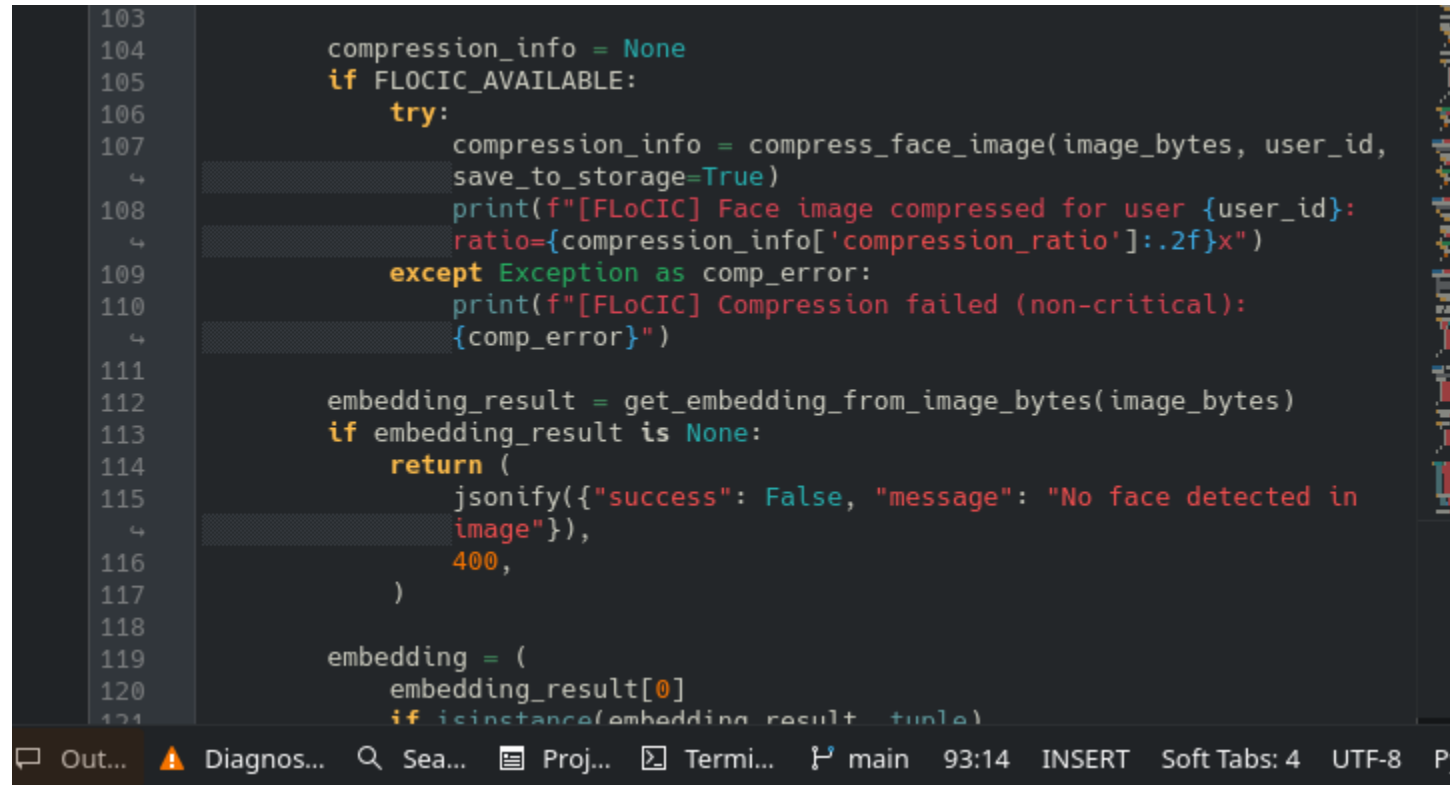
```

FLoCIC algoritem

FLoCIC algoritem: interpolativno kodiranje.

2. Integracija v Face Recognition API

```
103
104     compression_info = None
105     if FLOCIC_AVAILABLE:
106         try:
107             compression_info = compress_face_image(image_bytes, user_id,
108             ↪ save_to_storage=True)
109             print(f"[FLoCIC] Face image compressed for user {user_id}:
110             ↪ ratio={compression_info['compression_ratio']:.2f}x")
111         except Exception as comp_error:
112             print(f"[FLoCIC] Compression failed (non-critical):
113             ↪ {comp_error}")
114
115     embedding_result = get_embedding_from_image_bytes(image_bytes)
116     if embedding_result is None:
117         return (
118             ↪ jsonify({"success": False, "message": "No face detected in
119             ↪ image"}),
120             ↪ 400,
121         )
122
123     embedding = (
124         ↪ embedding_result[0]
125         ↪ if isinstance(embedding_result, tuple)
```



Integracija v API

Klic kompresije ob registraciji uporabnika v face_recognition_api.py.

3. API odgovor s podatki o kompresiji

```
1  {
2    "success": true,
3    "message": "User matic registered successfully",
4    "embeddings_count": 8,
5    "compression": {
6      "ratio": 2.47,
7      "original_size": 17462,
8      "compressed_size": 7068
9    }
10 }
11 |
```

API odgovor

REST API vrne razmerje kompresije pri uspešni registraciji.

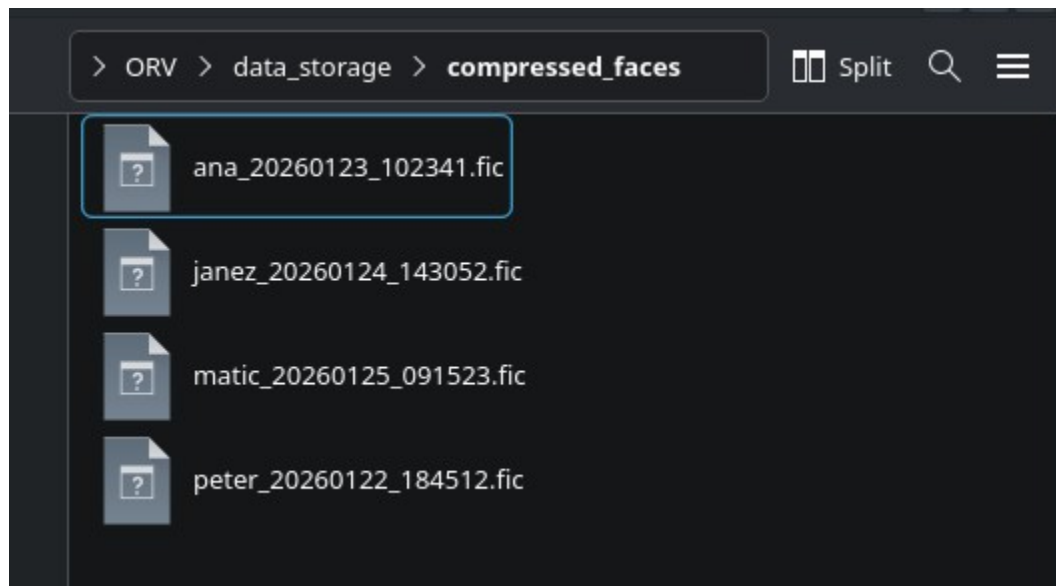
4. Shramba kompresiranih slik

```
1  {
2    "success": true,
3    "stats": {
4      "total_archived_images": 4,
5      "total_compressed_bytes": 28018,
6      "estimated_original_bytes": 69832,
7      "space_saved_bytes": 41814,
8      "average_compression_ratio": 2.49
9    }
10 }
11 |
```

Kompresiran arhiv

Kompresirane slike (.fic) se shranjujejo v data_storage/compressed_faces/.

5. Statistika kompresije



Statistika

Endpoint /compression/stats vrne statistiko prihranjenega prostora.

Rezultati

- **Povprečno razmerje kompresije:** ~2-3x za sivinske slike obrazov 128x128
- **Brezizgubna kompresija:** Dekompresirana slika je identična originalu
- **Avtomatsko arhiviranje:** Vsaka registracija uporabnika shrani kompresirano sliko